

School of Computer Applications and Technology

Semester End Examination - Jan 2026

Masters of Computer Applications

Duration : 120 Minutes

Max Marks : 50

Sem I - E1PA104B - Database Management SystemGeneral Instructions*Answer to the specific question asked**Draw neat, labelled diagrams wherever necessary**Approved data hand books are allowed subject to verification by the Invigilator*

- 1) Define briefly DDL, DML, and DCL and give one example for each CO1(5)
- 2) Compare the File System approach with the Database Management System (DBMS). CO1(5)
Explain any five differences between them with respect to:
Data redundancy, Data consistency, Data sharing, Security, Data independence
- 3) Differentiate between First Normal Form (1NF) and Second Normal Form (2NF). CO2(5)
- 4) Explain the Entity-Relationship (ER) Diagram used in database design. CO2(7)
Draw an ER diagram for a University Database System that maintains information about Students, Courses, and Departments with the following requirements:
-Each Student has a unique RollNo, Name, and Age.
-Each Department has a unique DeptID and DeptName.
-Each Course has a unique CourseID, CourseName, and Credits.
-A Student belongs to one Department, but a Department can have many Students.
-A Student can enroll in many Courses, and a Course can have many Students.
- 5) Compare and analyze the behavior of INNER JOIN vs OUTER JOIN in a given dataset . CO3(7)
Use suitable example to justify your answer.
- 6) Consider the following relation: CO3(7)
STUDENT (RollNo, StudentName, CourseID, CourseName, Instructor)
Functional Dependencies:
RollNo → StudentName, CourseID, CourseID → CourseName, Instructor
(a) Identify the transitive dependency. (b) Normalize the relation to 3NF. (c) Write the resulting relations.
- 7) In a banking system, two transactions are executed concurrently: T₁: Transfer ₹5,000 from Account A to Account B T₂: Compute total balance of Account A and Account B Due to interleaving, T₂ reads the updated balance of Account A but the old balance of Account B. Answer the following Questions based on above situation a) Is the schedule serializable? Explain. b) Which serial order would preserve correctness? c) What type of inconsistency is observed? CO4(14)

OR

Consider the following database schema:

STUDENT (RollNo, Name, DeptID, Age)

DEPARTMENT (DeptID, DeptName, HOD)

Primary Keys are mentioned in bold letters.

DeptID in STUDENT is a foreign key referencing DEPARTMENT.

CO4(14)

(A) Write SQL Queries for the following:

- Display the names of all students who belong to the Computer Science department.
- List the RollNo and Name of students whose age is greater than 20.
- Find the department name and HOD of the student having RollNo = 105.
- Display the names of students who belong to the department headed by 'Dr. Sharma'.

(B) Write Relational Algebra Expressions for the following:

- Retrieve the names of students who belong to the Information Technology department.
- Find the Natural Join of both tables.
- List the RollNo and Name of students who do not belong to any department.